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Problem 1

1.

$$\forall x(Seal(x) \rightarrow Swim(x))$$

2.

$$\forall x(Fish(x) \rightarrow \neg Fly(x))$$

3.

$$\forall x \exists y (Fish(y) \wedge Fly(y) \wedge (Fish(x) \wedge (x \neq y) \rightarrow \neg Fly(x)))$$

4.

$$\exists x \exists y \forall z (x \neq y \wedge Fish(x) \wedge Fly(x) \wedge Fish(y) \wedge Fly(y) \wedge ((Fish(z) \wedge z \neq x \wedge z \neq y) \rightarrow \neg Fly(z)))$$

Problem 2

Given

$$\forall x([(x < 0) \wedge D(x)] \rightarrow P(x))$$

Since x is positive integer, $(x < 0)$ is false:

$$\forall x([F \wedge D(x)] \rightarrow P(x))$$

Complement law:

$$\forall x(F \rightarrow P(x))$$

Thus:

$$\forall x(T)$$

Thus:

$$T$$

It is true.

Problem 3

$$\forall x \forall y (G(x, y) \rightarrow C(x))$$

Problem 4

The original statement is

$$\forall g \exists w (B(g, w) \geq 5)$$

The negation is

$$\exists g \forall w (\neg B(g, w) < 5)$$